

SUBODH KUMAR SAHU

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EDUCATION

B.E. in Computer Engineering

Terna Engineering College, Navi Mumbai

2022-26

CGPA: 8.26/10

Higher Secondary Certificate (HSC)

Atomic Energy Central School No. 4

2022

75.5%

TECHNICAL SKILLS

Core CS: DSA, OOPS, DBMS, OS, Computer Networks, SDLC

Languages: Python, C++, C, SQL

Web & Cloud: React.js, Next.js, Flask, FastAPI, AWS, DigitalOcean

AI & Data: LLMs (Openai,Groq,Qwen,Lamma), WebLLM, MCP, RAG, Agentic AI, LLMOps, LangChain, LangGraph, Agno

Tools & DBs: Linux, PyTorch, MongoDB, Git, GitHub Actions (CI/CD), Nginx

EXPERIENCE

Software Development

May 2025 – July 2025

B.A.R.C, Trombay

- Robotics Control System:** Developed control software for a **Bionic Robotic Hand** using **C++**, implementing motion control algorithms and kinematic routines for precise robotic manipulation.
- Desktop UI & Hardware Integration:** Designed a cross-platform control application using **PySide6 (Qt)** and engineered a **UART** communication bridge for reliable real-time interaction with the **Kinova Robotic Arm**.
- Embedded Systems & Automation:** Built real-time robotics software integrating hardware interfaces, serial communication, and automation workflows, enabling low-latency robotic control and device synchronization.
- Technical Implementation:** Applied **C++**, **PySide6**, **UART**, and embedded systems concepts to deliver a portfolio-grade robotics solution focused on hardware-software integration and real-time control.

PROJECTS

T5 Text Summarizer

FastAPI, Hugging Face Transformers, T5, PyTorch, NLP, REST API

- Designed and implemented** an NLP-based abstractive text summarization system using the **T5 Transformer** architecture, leveraging Hugging Face Transformers for sequence-to-sequence learning and text generation.
- Optimized** the inference pipeline through efficient preprocessing, tokenization, **beam-search decoding**, and hardware-aware execution (Apple MPS, CUDA, CPU).
- Integrated FastAPI REST APIs** with an interactive frontend to deliver real-time AI-powered summarization, showcasing scalable backend development and model serving.

Research Paper RAG Assistant

LangChain, ChromaDB, OpenAI API, Groq API, Hugging Face

- Engineered** an end-to-end **RAG pipeline** for conversational research paper analysis, implementing PDF ingestion, recursive text chunking, embedding generation, vector indexing, semantic retrieval, and LLM-powered answer generation.
- Designed** a semantic search pipeline using **ChromaDB** as the vector database for embedding storage and similarity-based retrieval.
- Integrated OpenAI GPT, Groq APIs, and LangChain** to enable multi-model inference, prompt engineering, and retrieval-augmented responses.

AI Adaptive Cyber Honeypot

Python, FastAPI, Docker, PostgreSQL, React, Nginx, Qwen 3.5 7B

- Engineered** an AI-driven cyber deception platform leveraging the **Qwen 3.5 7B** LLM for malicious payload classification, attacker intent analysis, and adaptive response generation.
- Designed** attacker behavior analytics through **session tracking** and interaction profiling to enhance threat intelligence and intrusion analysis.
- Deployed** a containerized **FastAPI** microservices architecture using Docker, PostgreSQL, and Nginx as a reverse proxy, with structured logging and a React dashboard.

CERTIFICATIONS

- Oracle:** AI Foundations Associate | Generative AI Professional | AI Vector Search Professional | Data Science Professional
- Google Cloud & AI:** Career Launchpad | Python AI Assistant | Introduction to NLP | Machine Learning Algorithms

ACHIEVEMENTS

- Smart India Hackathon (SIH) 2024 & 2025:** Institute Finalist, developing AI-powered solutions for real-world problem statements in India's largest innovation hackathon.
- Research Paper(IEEE Manuscript):** Ensemble Learning Approaches for Loan Approval Prediction: A Three-Phase Machine Learning Framework
- Winner-Special Mention Award:** Operation Trinetra Hackathon, Anna University (Top Tier out of 170+ teams).
- E-Cell Member:** Part of Technical and Web Development Team.
- Freelance Web Developer & Video Editor:** Developed responsive client websites and created high-quality promotional, branding, and social media video content.